

Symbol	Parameter	Conditions	Min	Typ	Max	Unit	
I _{DDA} (SCALER)	Scaler static consumption from V _{DDA}	BRG_EN=0 (bridge disable)	-	0.2	0.3	μA	
		BRG_EN=1 (bridge enable)	-	0.82	1		
t _{START_SCALER}	Scaler startup time	-	-	140	250	μs	
t _{START}	Comparator startup time to reach propagation delay specification	High-speed mode	-	2	5	μs	
		Medium mode	-	5	20		
t _D ⁽²⁾	Propagation delay for 200 mV step with 100 mV overdrive	High-speed mode	-	50	80	ns	
		Medium mode	-	0.5	0.9	μs	
	Propagation delay for step > 200 mV with 100 mV overdrive only on positive inputs	High-speed mode	-	50	120	ns	
		Medium mode	-	0.5	1.2	μs	
V _{offset}	Comparator offset error	Full common mode range	-	-	±20	mV	
V _{hys}	Comparator hysteresis	No hysteresis	-	0	-	mV	
		Low hysteresis	-	10	-		
		Medium hysteresis	-	20	-		
		High hysteresis	-	30	-		
I _{DDA} (COMP)	Comparator consumption from V _{DDA}	Medium mode	Static	-	5	9	μA
			With 50 kHz ±100 mV overdrive square signal	-	6	-	
		High-speed mode	Static	-	70	110	
			With 50 kHz ±100 mV overdrive square signal	-	75	-	

1. Refer to [Section 5.3.5: Embedded voltage reference](#).
2. Evaluated by characterization - Not tested in production.

5.3.21 OPAMP characteristics

Table 59. OPAMP characteristics

Specified by design – not tested in production unless otherwise specified.

Symbol	Parameter	Conditions	Min	Typ	Max	Unit
V _{DDA}	Analog supply voltage Range	-	2.5	3.3	3.6	V
CMIR	Common Mode Input Range	-	0	-	V _{DDA}	
V _{OFFSET}	Input offset voltage	25°C, no load on output	-	-	±1.5	mV
		All voltages and temperature, no load	-	-	±3	
		All voltages and temperature, no load high speed mode	-	-	±4	
ΔV _{OFFSET}	Input offset voltage drift	-	-	±10	-	μV/°C
TRIMOFFSETP TRIMLPOFFSETP	Offset trim step at low common input voltage (0.1*V _{DDA})	-	-	0.8	-	mV
TRIMOFFSETN TRIMLPOFFSETN	Offset trim step at high common input voltage (0.9*V _{DDA})	-	-	0.8	-	
I _{LOAD}	Drive current	-	-	-	500	μA
I _{LOAD_PGA}	Drive current in PGA mode	-	-	-	270	
C _{LOAD}	Capacitive load	-	-	-	50	pF